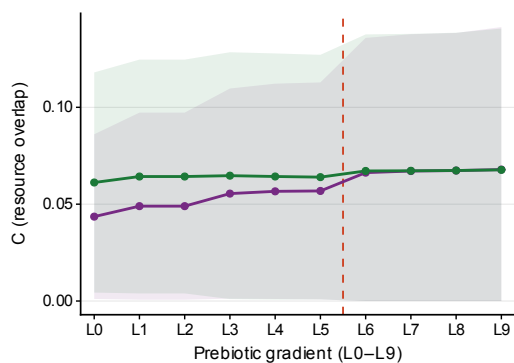


— *Akkermansia* × Gut — *Lactobacillus* × Gut

line = mean across viable pairs; shading (A, E, F) = $\pm$ SD

A

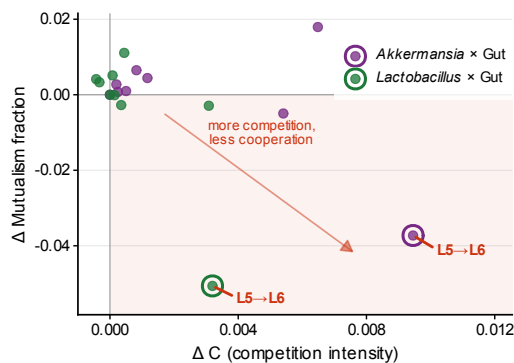
Competition intensity



B

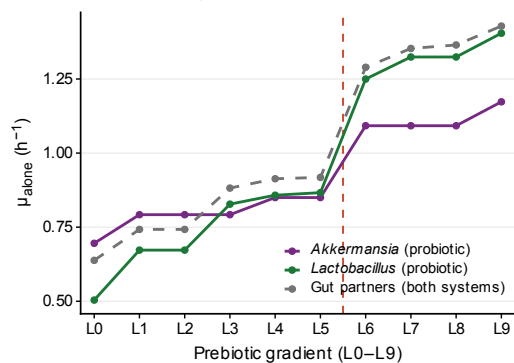
Δ Mutualism vs Δ Competition

Each point = one gradient step (L $\rightarrow$ L+1); rings = L5 $\rightarrow$ L6 (free glucose enters)



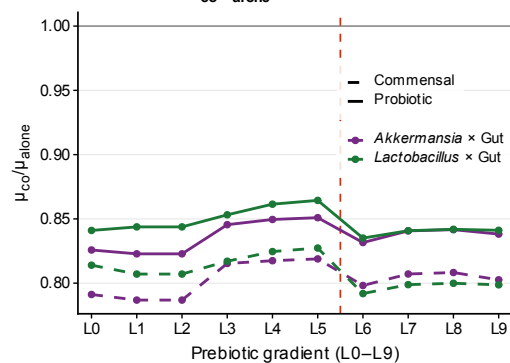
C

Monoculture growth rates



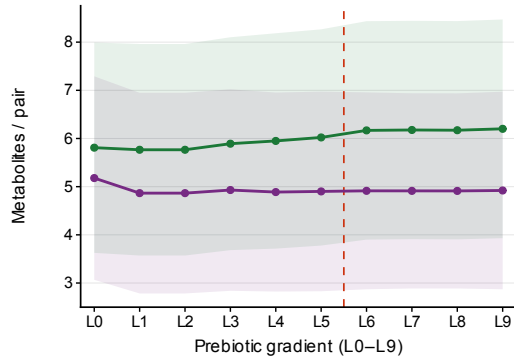
D

Growth ratio $\mu_{\text{co}}/\mu_{\text{alone}}$



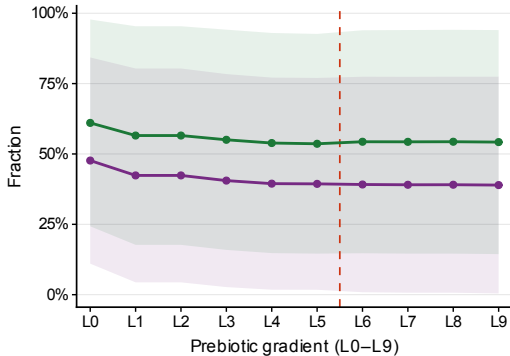
E

Cross-fed metabolites



F

Gene-supported fraction



G

Controlled contrast (richness fixed): pectin $\rightarrow$ glucose lowers mutualism in both systems

